WEEK 03 • IS321 ENTERPRISE ARCHITECTURE

discusses how EA can be AI-powered

The EA Approach: Core Components

The Five Levels That Structure Every Enterprise Architecture — and How AI Is Changing Enterprise Decision-Making

ROADMAP

What We'll Cover

01	The EA Approach	A structured way to organize enterprise complexity into consistent, documentable levels
02	Level 1: Goals & Initiatives	The strategic layer — why the enterprise exists and where it's headed
03	Level 2: Products & Services	What the enterprise delivers to customers or citizens
04	Level 3: Data & Information	The facts the enterprise needs to operate and decide
05	Level 4: Systems & Applications	The software that creates, moves, and stores that information
06	Level 5: Networks & Infrastructure	The hardware, networks, and platforms everything runs on
07	How the Levels Connect	A worked example showing change ripple top-down through all five
08	AI-Driven Enterprise Decision Support	Strategic decision-making, process optimization, and predictive analytics

What Is the EA Approach?

> *The EA Approach organizes everything an enterprise does — from strategy down to the wiring in the server room — into five consistent levels, so the whole enterprise can be documented, compared, and changed using one common structure.*

— Core idea behind the EA component levels (Bernard, 2012)

Why organize EA this way?

One Structure, Whole Enterprise

implement across the entire enterprise

Every part of the enterprise — no matter how different — fits into the same five levels.

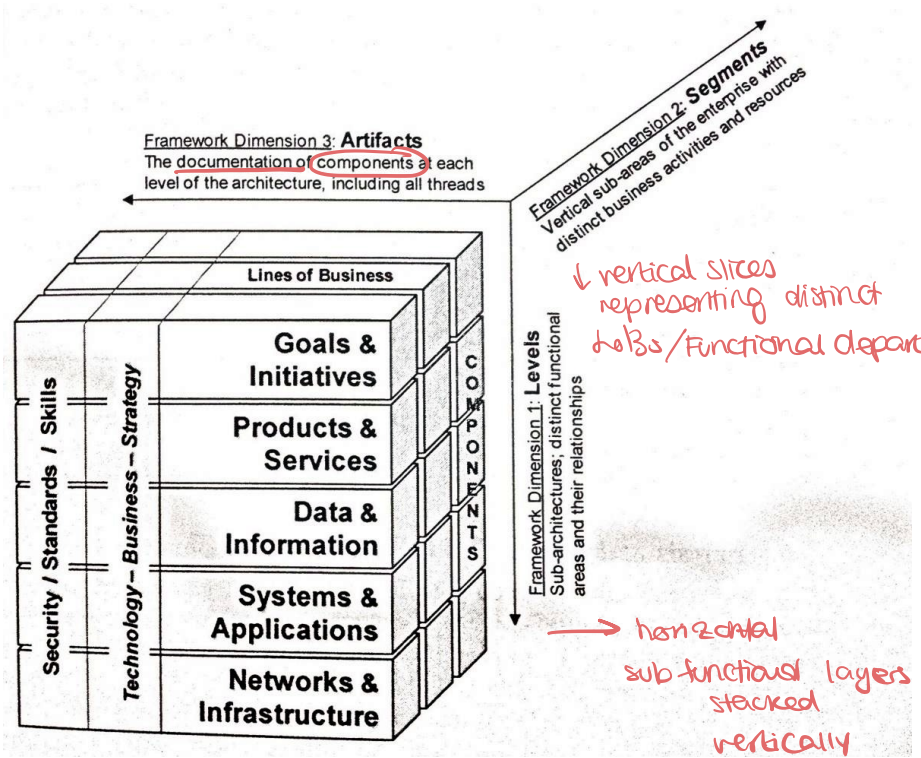
Traces Cause and Effect

A change entered at one level (say, a new strategic goal) can be traced down to exactly which systems and infrastructure it touches.

A Shared Vocabulary

Executives, business owners, and IT staff can all point to the same level when discussing a problem.

Five Levels, Top to Bottom



— Bernard (2012), An Introduction to Enterprise Architecture

The EA³ Cube Framework™

Bernard's original framework figure — five hierarchical levels, with Segments and Artifacts as the other two dimensions, and Security/Standards/Skills as a thread running through every level.

LEVEL 1 OF 5

Goals & Initiatives

defines a company's long-term strategic plans



The strategic layer

Defines the enterprise's mission, strategic goals, and the initiatives (programs and projects) meant to achieve them. Everything else in the EA — every product, system, and server — should trace back to a goal at this level.

EXAMPLE

A university sets the strategic goal "expand access to remote learners" and launches an initiative called "Online Degree Expansion" to accomplish it.

What lives at this level

- Mission and vision statements
- Strategic goals (multi-year targets)
- Strategic initiatives (programs/projects)
- Outcome measures that define success

LEVEL 2 OF 5

Products & Services



The delivery layer

Defines what the enterprise actually produces and delivers — the products, services, and business processes that customers, citizens, or internal units experience directly. This is where strategic goals become tangible offerings.

EXAMPLE

To deliver on "Online Degree Expansion," the university creates a new product: a fully online bachelor's degree track, plus supporting services like virtual advising.

What lives at this level

- Customer-facing products and services
- Internal business processes
- Business process models and workflows
- Service delivery channels

LEVEL 3 OF 5

Data & Information



The knowledge layer

each level / layer supports / provides a backbone to upper levels

Defines the facts, records, and information flows the enterprise needs to deliver its products and services and to make decisions — independent of which system happens to store them.

EXAMPLE

The online degree program needs student enrollment records, course-completion data, and advising notes — defined once, used consistently across every system that touches them.

What lives at this level

- Data models and entity definitions
- Information exchanges between systems
- Data quality and stewardship rules
- Reporting and analytics data needs

LEVEL 4 OF 5

Systems & Applications



The processing layer

Defines the software applications and systems that create, store, process, and move the enterprise's data — the tools that make the products/services at Level 2 actually work day to day.

EXAMPLE

A Learning Management System (LMS), a student information system, and a virtual-advising portal are the applications that run the online degree program.

What lives at this level

- Enterprise and departmental applications
- System interfaces and integrations
- Application architecture diagrams
- Software licensing and lifecycle

LEVEL 5 OF 5

Networks & Infrastructure



The foundation layer

hosts and executes L4 SW

Defines the hardware, networks, data centers, and cloud platforms that every system and application at Level 4 depends on to actually run.

EXAMPLE

The LMS and student portal run on cloud servers with enough bandwidth and storage to support thousands of remote learners streaming lectures simultaneously.

↳ high-capacity storage to accommodate learners

What lives at this level

- Servers, storage, and networks
- Cloud and data-center platforms
- Security and infrastructure standards
- Capacity and performance planning

Top-Down Planning, Bottom-Up Support

Top-Down: Strategy Drives Design

A new goal at Level 1 creates requirements for new products (L2), which need new data (L3), new systems (L4), and new infrastructure (L5). This is the normal, preferred direction of EA planning.

L1 → L2 → L3 → L4 → L5

Requirements flow downward

Bottom-Up: Technology Can Enable Change

Sometimes a new technology (L5) makes a new system possible (L4), which enables new information flows (L3), new services (L2), and even new strategic goals (L1) — e.g., cloud computing enabling entirely new business models.

L5 → L4 → L3 → L2 → L1

Valid, but should stay the exception

WORKED EXAMPLE

Amanah Bank Sets a New Strategic Goal

Amanah Bank decides to pursue open banking compliance. Watch how that single goal ripples down through every level.



The whole chain must also be documented consistently across every affected segment — and stay within existing security and standards constraints (a cross-cutting thread, not a sixth level).

NEW SECTION

Modernizing EA & phasing out legacy systems

AI-Driven Enterprise Decision Support

The five-level model tells us what to document. AI is changing how enterprises use that documentation — turning it from a static reference into an active decision-making tool.

→ traditionally EA documentation resided in static PDF- repositories

op 1

Strategic Decision-Making

op 2

Business Process Optimization

op 3

Predictive Analytics for Planning

Valerts you about the future

AI for Strategic Decision-Making



AI helps analyze enterprise systems and improve strategic decision-making by surfacing patterns across the five levels that a human reviewing documentation manually would likely miss — for example, spotting that three Level-2 products all depend on the same aging Level-4 system.

EXAMPLE

Feeding an EA repository's Level 1 goals and Level 2 product list into a generative AI tool (per PSU's approved ChatGPT/Gemini policy) to ask: "Which of our current products best support the goal of open banking compliance, and which are unrelated legacy offerings we could retire?"

AI for Business Process Optimization

AI supports business process optimization by analyzing how work actually flows through the Products & Services and Systems & Applications levels, then recommending where steps can be automated, reordered, or eliminated.

EXAMPLE

The student-advising process for the new online degree involves five manual handoffs between systems. An AI-assisted process-mining tool flags that two of those handoffs could be merged into a single automated step in the LMS, cutting advising turnaround time.

reduces handoffs to optimize the workflow and reduce delay

Predictive Analytics for Enterprise Planning

Predictive analytics uses historical Data & Information (Level 3) to forecast future demand on Systems & Applications (Level 4) and Networks & Infrastructure (Level 5) — letting the enterprise plan capacity before it becomes a problem, not after.

to revert to servers with higher capacity beforehand through predictive levels analytics powered by AI

EXAMPLE

Enrollment data from Level 3 shows online-degree sign-ups growing 40% each semester. A predictive model projects that the current cloud hosting plan (Level 5) will hit capacity limits within two semesters — giving IT leadership time to plan the upgrade instead of reacting to an outage.

SYNTHESIS

AI Applied Across the Five Levels



L1 Goals

AI flags which strategic goals lack supporting products or are under-resourced.



L2 Products

AI recommends process optimizations within product delivery.



L3 Data

AI identifies data quality gaps that would undermine predictive models.



L4 Systems

AI surfaces redundant or conflicting applications across lines of business.



L5 Infrastructure

Predictive analytics forecasts capacity needs before limits are hit.

SUMMARY

Key Takeaways

- The EA Approach organizes every enterprise into five consistent levels: Goals & Initiatives, Products & Services, Data & Information, Systems & Applications, and Networks & Infrastructure.
- Requirements normally flow top-down (strategy → infrastructure), though technology can occasionally enable bottom-up change.
- A single strategic goal can be traced all the way down to the specific infrastructure it requires — as shown in the Amanah Bank example.
- AI-Driven Enterprise Decision Support adds three capabilities on top of this model: strategic decision-making, business process optimization, and predictive analytics for planning.
- Any AI tool use must follow PSU's AI-and-EA-Tools policy — acknowledged and subject to academic-integrity requirements.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.) — Primary text
- The Open Group — TOGAF Standard, Version 9.2

Next: Week 4

EA as a Management Program — the EA Management Plan's four parts. Mapped to CLO1, CLO3.

Questions?